

9/17/26

Chapter 1: Sequence Alignment Algorithms

Learning Goal: Learn about the combinatorics of sequences (prefixes, suffixes, etc.) and how dynamic programming allows all of this to work

1.1 Edit Graph & Alignment

1.2 Scoring Alignment, Scoring Scheme

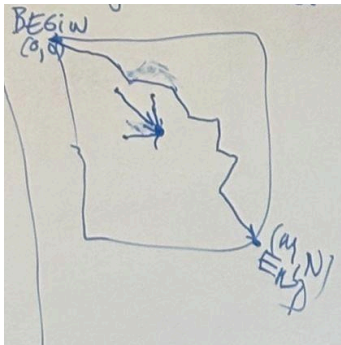
	A	T	C	G
A	1	0	0	0
T	0	1	0	0
C	0	0	1	0
G	0	0	0	1

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unit scoring scheme

- $X = x_1, x_2, \dots, x_M$
- $Y = y_1, y_2, \dots, y_N$
- $x_i, y_j \in \Sigma = \{A, C, G, T\}$
- $\delta(x_i, y_j), \delta(-, y_j), \delta(x_i, -)$
- Ex:
 - $X = \{A \ T \ T \ A \ C \ G\}$
 $\quad \quad \quad | \ | \ | \ | \ | \ |$
 - $Y = \{A \ T \ A \ T \ C \ G\}$
 - Score 4
 - Optimal score
 - $X = \{A \ T \ T \ A \ - \ C \ G\}$
 $\quad \quad \quad | \ | \ | \ | \ | \ |$
 - $Y = \{A \ - \ T \ A \ T \ C \ G\}$
 - Score 5
- $G = (V, E)$, V = vertex set, E = edge set
- Edit Graph $G_{X,Y}$: edge-labeled directed graph
- Vertices $(M+1) * (N+1)$; start numbering from 0
 - (i,j) row i , col j
- Edges:
 - Gap in Y
 - $1 \leq i \leq M$
 - $1 \leq j \leq N$
 - $\begin{pmatrix} x_i \\ - \end{pmatrix}$
 - $(i-1, j) \rightarrow (i, j)$

- Above the arrow is the label of the edge
- Gap in X
 - $1 \leq i \leq M$
 - $1 \leq j \leq N$
 - $$\begin{pmatrix} - \\ y_j \end{pmatrix}$$
 - $(i, j-1) \rightarrow (i, j)$
- Alignment edge
 - $1 \leq i \leq M$
 - $1 \leq j \leq N$
 - $$\begin{pmatrix} x_i \\ y_j \end{pmatrix}$$
 - $(i-1, j-1) \rightarrow (i, j)$
- **Conclusion:** Only 3 types of edges coming to (i, j)
- **Theorem:** There is a one-to-one correspondence between directed paths from $(0, 0)$ to (M, N) and alignments between X and Y



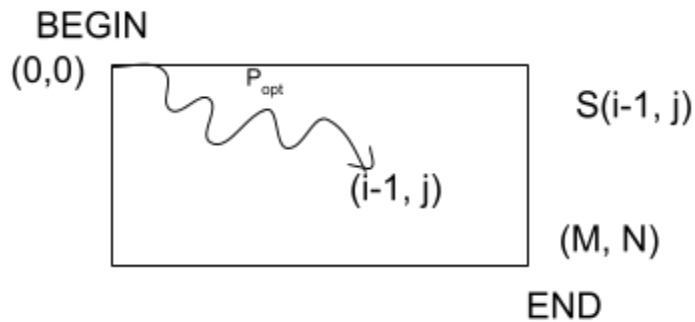
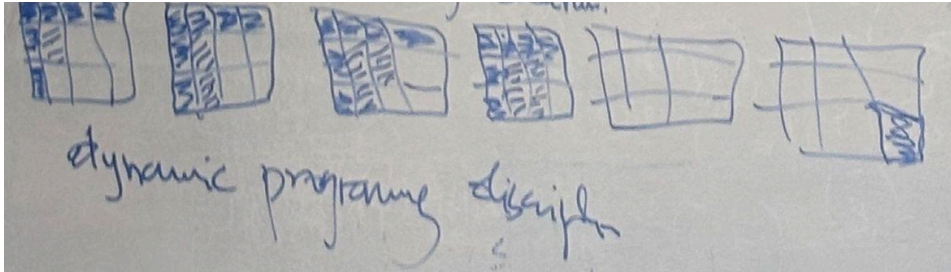
- **Directed Acyclic Graph**

Dynamic Programming + Principle of Optimality

- Def: Denote $S(i, j)$ as the score of maximum path from BEGIN to (i, j)
- Because there are only 3 edges directed into vertex (i, j) , it follows that any optimal(max) path P_{opt} to (i, j) must use one of these edges:
 - P_{opt} first arrives at $(i-1, j)$ and it follows a gap-in-Y edge into (i, j)
 - P_{opt} first arrives $(i, j-1)$ and then through gap-in-X edge to (i, j) δ
 - P_{opt} get to $(i-1, j-1)$ and through an alignment edge gets to (i, j)
- Recall: $S(i, j) = \text{MAX of } \{$

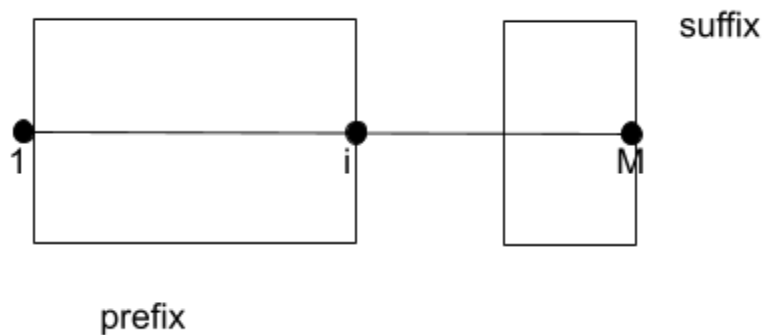
$$\begin{aligned} &S(i-1, j-1) + \delta(x_i, y_j), \\ &S(i-1, j) + \delta(x_i, -), \\ &S(i, j-1) + \delta(-, y_j) \} \end{aligned}$$
- We need to compute $S(i, j)$ such that at any time we have already computed the values of S needed by the recurrence (at least one of the indices i and j smaller)

- There are several ways to solve such a computation; we will do it column by column (see below)



Global Alignment Algorithm

- Useful for detecting things like plagiarism
- It is a quadratic algorithm, so order of complexity is $O(M*N)$ because you need to compute the entire matrix
 - But it's still ultimate simplicity, given only 3 lines essentially
 - If $M=N$, $O(N^2)$



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- $S[0,0] = 0$
- FOR $j=1$ to N DO
 - $S[0,j] = S[0,j-1] + \delta(-, y_j)$
- FOR $i=1$ to M DO
 - $S[i,0] = S[i-1,0] + \delta(x_i, -)$
 - FOR $j = 1$ TO N DO

$$S[i,j] = \text{MAX} \begin{cases} [S[i-1,j-1] + \delta(x_i, y_j)] \\ [S[i-1,j] + \delta(x_i, -)] \\ [S[i,j-1] + \delta(-, y_j)] \end{cases}$$

PRINT("MAX Score", S[M,N])

- Ungapped multiple alignment
- WWYIR
- WFYVR
- WYYVR
- WYYTR
- WFKR
- WYYVR

- BABA
- AAAC
- AACC
- AABA
- AACC
- AABC

- There are 24 amino acids OBSERVED
 - 14 are A, 4 are B, 6 are C
 - The observed proportions: A=14/24, B=4/24, C=6/24
 - Pairs of letter aligned to each
 - $4 * 6 \text{ choose } 2 = 60 \text{ pairs aligned}$

Aligned Pair	Frequency Observed
A to A	26/60
A to B	8/60
A to C	10/60
B to B	3/60
B to C	6/60
C to C	7/60

Compare observed frequencies of the expected frequencies

- We will generate random sequences using the frequencies of aa (A 14/24, B 4/24...). That is, we chose at random sequences with same frequency of aa given above, and make an alignment of them. In a random alignment the EXPECTED proportion of pairs is
 - A to A $(14/24) * (14/24)$ (independence)
 - A to B $(14/24)*(4/24)$ (A to B or B to A)
 - A to C
- These fractions are now used to calculate the “estimated likelihood ratios”
 - $2*\log_2\left(\frac{observed}{expected}\right)$

Margaret Dayhoff table

Aligned Pair	Frequency Observed	Expected Frequency	$2*\log_2\left(\frac{observed}{expected}\right)$
A to A	26/60	196/576	0.70
A to B	8/60	112/576	-1.09
A to C	10/60	168/576	-1.61
B to B	3/60	16/576	1.79
B to C	6/60	48/576	0.53
C to C	7/60	26/576	1.83

	A	B	C
A	1	-1	-2
B	-1	2	1
C	-2	1	2